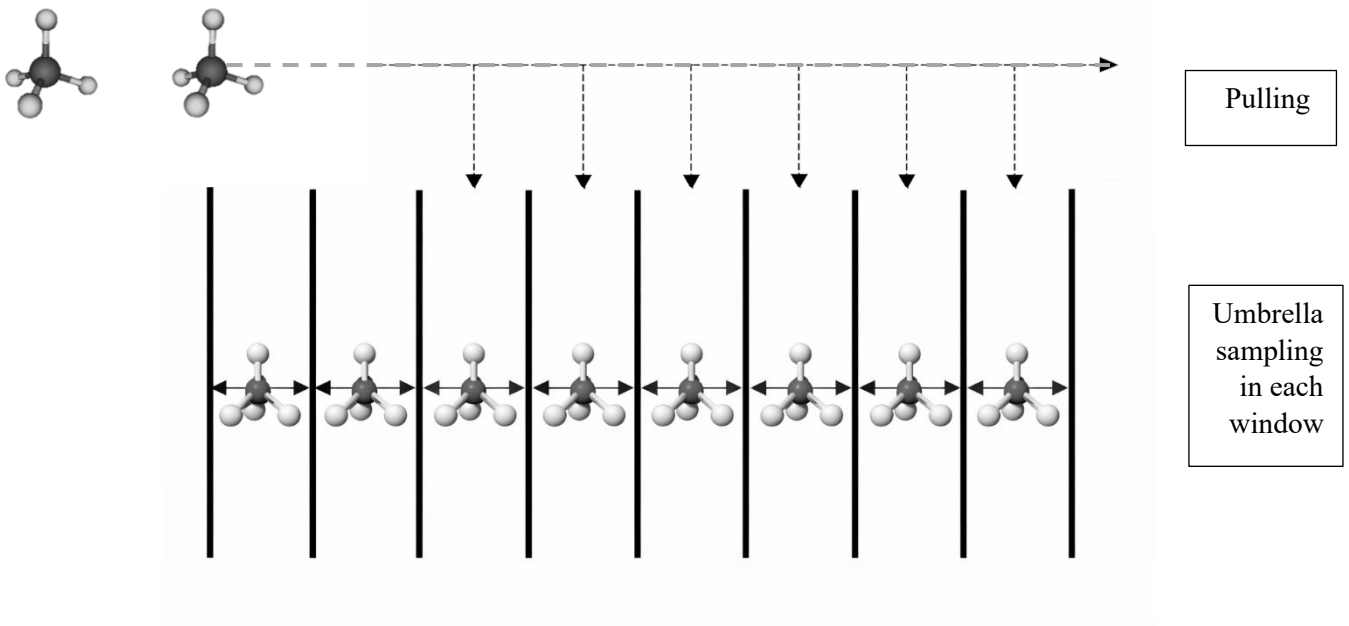
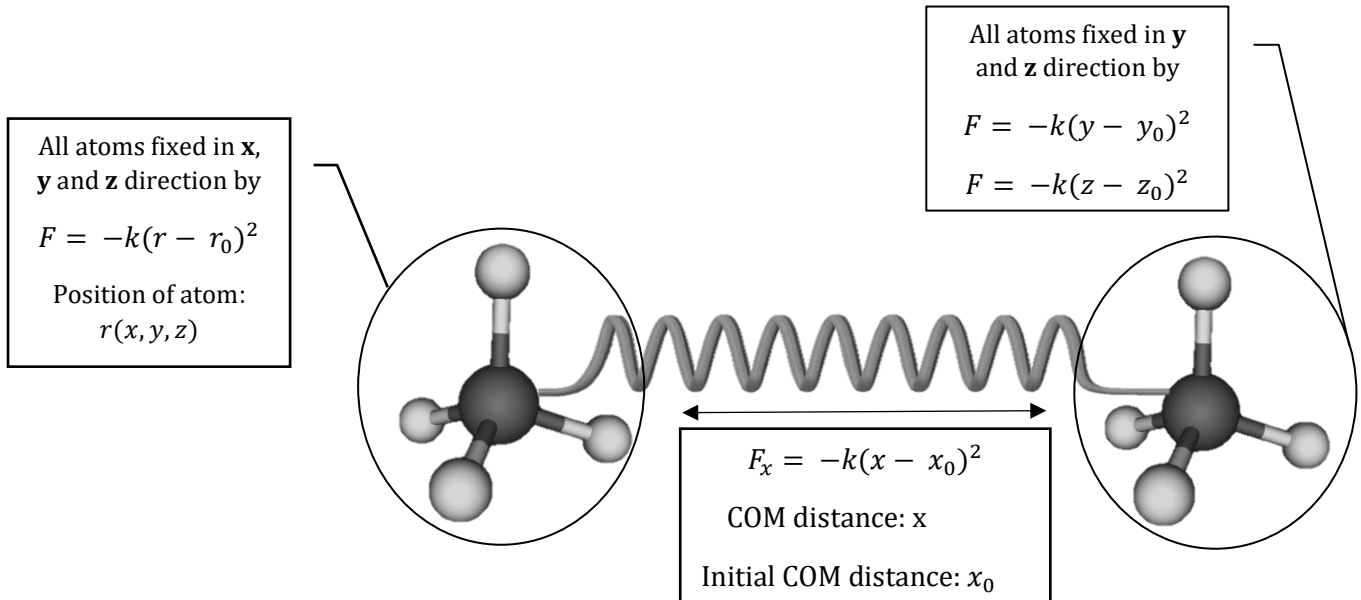


umbrella sampling schematic



Umbrella Sampling Procedure

Umbrella sampling is used to calculate the **free energy (PMF)** along a **reaction coordinate**. In our system, the reaction coordinate is the **center-of-mass (COM) distance between methane1 and methane2**, chosen along the **x-axis**.

step 1: Generating Starting Configurations (Pulling)

1. First, perform a **continuous pulling simulation**:
 - Methane2 is **pulled away from methane1** along the x-axis.
 - A **harmonic spring** connects the COMs of methane1 and methane2. (Methane1 is restrained in 3 directions, Methane2 is pulled in x-axis and restrained in z y)
 - The spring is stretched at a **constant velocity** (parameter --> pulling-rate).
 2. Important:
 - This pulling step is **NOT umbrella sampling**.
 - The harmonic potential only guides the system to generate configurations.
 3. Simulation setup:
 - Pulling-rate: e.g. 0.01 ps/nm
 - Total pulling time. (selected according to the length of the reaction coordinate and the size of the box).
 - Save structures frequently (e.g. every 1 ps).
-

step 2: Extracting Configurations for Umbrella Windows

1. After pulling is finished:
 - Extract **all saved frames** from the trajectory. (conf0.xyz, conf1.xyz, conf2.xyz, ...)
 - For each frame, calculate the **COM distance** between methane1 and methane2 and save in a summary_distance.dat file.
2. Choose a **window spacing**:
 - E.g. for our simulation 0.5 Å.
3. Identify frames whose COM distances match the desired spacing.(by user from summary_distance.dat)
E.g. Frame Distance (Å)

```
....  
6      3.5  
160    4.0  
173    4.5  
....
```

- Use `conf6.xyz`, `conf160.xyz` and `conf173.xyz`, ... as starting structures for three umbrella windows.
-

step 3: Preparing Umbrella Sampling Simulations

1. Each selected configuration becomes **one umbrella window**.
 - If 15 configurations are selected → **15 independent simulations**.
2. For umbrella sampling:
 - Use the **same harmonic potential** as in pulling.
 - **Set** pulling-rate = 0.
 - the spring does **not move**.
 - the system is **restrained around a fixed COM distance**.
 - the **initial COM distance** will be used as the **reference distance**.
3. Each window samples fluctuations around its target COM distance.